

Project Report — Predictive Analytics Using Historical Data

Overview This project demonstrates building simple predictive models to forecast sales using historical data. It includes data loading, preprocessing, baseline models (linear trend), a time-series model (ARIMA), evaluation, and visualizations.

Data
Source: ``data/sales_data.csv`` — expects a ``date`` column and a numeric sales column.

Preprocessing: daily resampling, forward-fill missing days, sort by date.
Methods
Linear trend regression: uses integer time index as regressor. Serves as a baseline.
ARIMA(1,1,1): simple autoregressive integrated moving average model as a stronger baseline.
Evaluation: MAE, RMSE, MAPE, and AIC (for ARIMA).
Visualization: actual vs predicted, prediction intervals, residuals distribution, rolling MAE.

Results (sample dataset) See

``report/Results_Summary.md`` for numeric metrics and interpretation.

Visual outputs are in ``notebooks/sales_forecasting.ipynb`` (plots generated by ``src/visualization.py``).

How to reproduce
1. Create a Python environment and install requirements:
`bash`python -m venv .venv`
venv\Scripts\activate # Windows`pip install -r requirements.txt``

2. Open ``notebooks/sales_forecasting.ipynb`` and run all cells (or run ``src/generate_report.py`` if added).

3. Replace ``data/sales_data.csv`` with your dataset (ensure ``date`` and ``sales`` columns).
Next steps
Hyperparameter tuning with ``pmdarima`` or grid search.
Add external regressors and seasonal decomposition.
Save model artifacts (``joblib``) and create a small pipeline to retrain and redeploy models periodically.